VIMARSH SATHIA **CS17B046**
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EDUCATION

Program	Institution	%/CGPA	Year
B.Tech, Computer Science	Indian Institute of Technology Madras	**9.23**/10	2021
Class XII (ISC)	Smt. Sulochanadevi Singhania School	95.2%	2016
Class X (ICSE)	Hiranandani Foundation School, Thane	96.3%	2014

RELEVANT COURSE WORK

Design & Analysis of Algorithms *(Theory+Lab)*	**Programming and Data Structures** *(Theory+Lab)*	**Graph Theory**
Computer Organization & Architecture *(Theory+Lab)*	**Foundations of Computer System Design** *(Theory+Lab)*	**Probability, Statistics & Stochastic Processes**

SKILLS

Programming languages: C++, C, Java, Python | **Software Packages:** MATLAB/Octave, cmake, Git

SCHOLASTIC ACHIEVEMENTS

- Awarded a **branch change** from Engineering Physics to Computer Science at the end of Semester 1 after securing a GPA of **9.75** *(2018)*
- Secured an All India Rank of **1806** (amongst 1,50,000+ candidates) in JEE Advanced 2017. *(2017)*
- Secured Governor's Gold Medal for scoring above 90% in all subjects undertaken in Class X exams . *(2014)*

PROFESSIONAL EXPERIENCE

Data Mining Intern, Actify Data Labs Guide: Mr. Bilal Arif *(June – July 2019)*

- Generated data using a training dataset, for modeling purposes.
- Built an approximate empirical copula from the given dataset, drew random samples from it and used inverse marginal column quantiles with linear interpolation to generate random vectors.
- Used hashing with chaining as an optimized data structure equivalent to n-dimensional sparse matrices in C++

Quantum Computing Intern, DXC Labs, Bangalore Guide: Mr. Ganesh Swaminathan *(June – July 2018)*

- Simulated and demonstrated standard quantum computing algorithms.
- Used the Q# programming language and IBM QISKit composer to code and simulate standard procedures.
- Demonstrated a simple Grover's search algorithm circuit to introduce an audience to concepts in Quantum Information and its advantages in particular situations.
- Featured in DXC Blogs for my work as a freshman here and here.

PROJECTS

Scalar and Superscalar Pipelined Processor Design Guide: Prof. Madhu Mutyam(IITM) *(Jan – May 2019)*

- Designed a scalar and superscalar pipelined processor that has a 256B instruction cache and a 256B data cache; both having a read-write port each with direct mapping (with block size 4B).
- The program employs a Reduced Instruction Set Computer (RISC) processor which achieves pipelining for instructions and minimizes the number of cycles per instruction to approximately 1.

Designing a Hack Computer Guide: Prof. Madhu Mutyam(IITM) *(Jul – Nov 2018)*

- Designed a Hack Computer Platform using the Hack Hardware Description Language(HDL) specified in the online tool nand2tetris.
- The computer consists of Memory, CPU, Instruction Memory and the integration of the three.

Bluetooth Controlled Cleaning Bot Center For Innovation(CFI), IIT Madras *(Nov 2017)*

- Built a cleaning bot with an Arduino, a central suction mechanism and remote control via a mobile app
- Secured a spot in the Asia Book of Records for the largest number of bots cleaning an area

POSITIONS OF RESPONSIBILITIY

Events Team, Saarang , IIT Madras. *(April 2018-March 2019)*

- Coordinated various events during the semester and Saarang like open mics and improv workshops as a coordinator at Comedy Club, IIT Madras.
- Revamped the Facebook page of Comedy Club and achieved a **150%** increase in social outreach.
- Monitored a budget of around **INR 50,000** for conducting the events.

EXTRA-CURRICULAR ACTIVITIES

Institute Squash Team, IIT Madras *(September 2018 – Present)*

- Represented the IITM squash contingent at the Inter IIT Sports Meet, 2018, held at IIT Guwahati
- Won a bronze medal for clinching the 3rd place at the sports meet.

Competitive Programming *(2018 – Present)*

- Regularly practice and participate in online coding competitions held on Codeforces. Handle: vimarsh6739